Mixed Integer Linear Programming

Some x's are integers where some are continuous variables.

Note that $LP \in P$ but ILP can be used to solve $NP - Complete$ problems so the integer constraints makes problem significantly different

Vertex Cover as Binary ILP (Integer Linear Programming)

Given a a graph $G = (V, E)$ find $\min |V'|, V' \subset V$ where V' covers all the edges.

$x_i = 1 \leftrightarrow i \in V'$ else 0

$\min \sum x_i$

s.t. $x_i + x_j \geq 1(\{i, j\} \in E)$

$\forall i(i \in V) 0 \leq x_1 \leq 1$

$\forall i(i \in V) x_i$ integer